INITIALIZE ELEMENTS IN AN ARRAY

```
#include <stdio.h>
int main()
{
int SIZE,element;
printf("Enter the size of array:\n");
scanf(“%d\n”,&SIZE)
int numbers[SIZE];
 for (int i = 0; i < SIZE; i++)
  {
      printf(“enter the elements:\n”);
      scanf(“%d”,&element);
       numbers[i] = element;
 }
printf("Initialized array elements:\n");
   for (int i = 0; i < SIZE; i++)
       printf("numbers[%d] = %d\n", i, numbers[i]);
 return 0;
}
```

FINDING SMALLEST AND LARGEST IN ARRAY

```
#include <stdio.h>

int main() {
   int arr[100];
   int size, i;
   int max, min;

   // Prompt user for array size
   printf("Enter the number of elements in the array: ");
   scanf("%d", &size);

   // Input array elements
   printf("Enter %d elements:\n", size);
   for (i = 0; i < size; i++) {
      scanf("%d", &arr[i]);
   }
   // Initialize max and min with the first element
   max = arr[0];
   min = arr[0];
   // Traverse the array to find largest and smallest
   for (i = 1; i < size; i++) {
      if (arr[i] > max) {
         max = arr[i]; // Update max if current element is larger
      }
      if (arr[i] < min) {
         min = arr[i]; // Update min if current element is smaller
      }
```

```
    }

    // Output the results
    printf("Largest element = %d\n", max);
    printf("Smallest element = %d\n", min);

    return 0;
}
```

BUBBLE SORT

```
#include <stdio.h>

int main() {
    int arr[100], n, i, j, temp;

    // 1. Take array size input
    printf("Enter number of elements: ");
    scanf("%d", &n);

    // 2. Take array elements input
    printf("Enter %d integers:\n", n);
    for (i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    // 3. Bubble Sort Algorithm (Ascending Order)
    for (i = 0; i < n - 1; i++) {
        for (j = 0; j < n - 1 - i; j++) {
            if (arr[j] > arr[j + 1]) { // Change '>' to '<' for Descending order
                // Swap elements
                temp = arr[j];
                arr[j] = arr[j + 1];
                arr[j + 1] = temp;
            }
        }
    }

    // 4. Print the sorted array
    printf("Sorted array in ascending order:\n");
    for (i = 0; i < n; i++) {
        printf("%d ", arr[i]);
    }
    printf("\n");

    return 0;
}
```

SEARCHING ELEMENTS IN ARRAY

```
#include <stdio.h>

// Function to perform linear search
int linearSearch(int arr[], int size, int target) {
    for (int i = 0; i < size; i++) {
        // Check if the current element matches the target
        if (arr[i] == target) {
            return i; // Return the index if found
        }
    }
    return -1; // Return -1 if the element does not exist
}

int main() {
    int arr[100], size, target, result;

    // Get the number of elements from the user
    printf("Enter the number of elements in the array: ");
    scanf("%d", &size);

    // Input the array elements
    printf("Enter %d integers:\n", size);
    for (int i = 0; i < size; i++) {
        scanf("%d", &arr[i]);
    }

    // Get the target value to search for
    printf("Enter the element to search for: ");
    scanf("%d", &target);

    // Call the search function
    result = linearSearch(arr, size, target);

    // Output the result
    if (result != -1) {
        printf("Element %d found at index %d (Position %d).\n", target, result, result + 1);
    } else {
        printf("Element %d not found in the array.\n", target);
    }

    return 0;
}
```